

ISLAM MANSOUR

APPLIED DATA SCIENCE STUDENT | MACHINE LEARNING & DATA ENGINEERING

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Applied Data Science student ranked 6th in the degree program at TU Hamburg. Builds end-to-end products across machine learning, ETL, backend APIs, media processing and interactive analytics, with a focus on reliable implementation, validation and clear business-facing outputs.

EDUCATION

B.Sc. Applied Data Science - Hamburg University of Technology (TU Hamburg) | *Ranked 6th in the degree program 2022 - 2026*

Focus: machine learning, statistics, databases and data engineering. Relevant coursework includes Big Data, algorithms and data structures, and statistics.

SELECTED PROJECTS

Production-Style Churn Prediction API | *Python | Scikit-learn | FastAPI | Pydantic*

Deployed a trained gradient-boosting model as a documented REST API. Added schema validation, health-check endpoints, batch prediction and model serialization.

ETL Pipeline & Data Warehouse | *Python | SQL | Pandas | SQLite*

Integrated CSV, JSON and SQLite sources into staging and mart layers. Implemented CTEs, window functions, automated SQL views and six passing data-quality tests covering nulls, uniqueness and referential integrity.

Interactive Sales Dashboard | *Streamlit | Plotly | SQLite | Flask*

Built a dashboard with cached SQL access, KPI cards and live filters for region, month and category.

Chess API with AI | *FastAPI | Pydantic | SQLite | Minimax | Stockfish*

Implemented legal-move validation, persistent PvP games, move history and selectable minimax, random or Stockfish opponents.

Git-Style Repository API | *REST API | Push/Pull Workflows*

Created API-driven push and pull workflows for transferring and synchronizing project content.

Media Optimization API | *REST API | Image & Video Processing*

Built automated upload and transformation endpoints that optimize images and selected video formats.

Virtual GPU Renderer | *Python | NumPy | Rasterization | Shaders*

Implemented a software GPU pipeline with programmable shaders, depth testing, textures and Phong lighting.

Data Cleaning & EDA | *Python | Pandas | Seaborn | Matplotlib*

Resolved missing values, duplicates, category inconsistencies and outliers in real-estate data, then produced statistical and visual analysis.

Additional projects: NLP sentiment analysis, four-model churn benchmark, automated PDF reporting, C++ ray tracing, animated 3D chess and C terminal games.

TECHNICAL SKILLS

Programming: Python, SQL, C++, C

Data & ML: Pandas, NumPy, Scikit-learn, Gradient Boosting, Random Forest, NLP, TF-IDF, Matplotlib, Seaborn, Plotly

Engineering & Tools: REST APIs, ETL pipelines, Pydantic, SQLite, FastAPI, Flask, Streamlit, Git, media processing, software rendering, Linux

LANGUAGES

Arabic: Native | **German:** Fluent | **English:** Fluent